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Unit 5 : Session 6 : SLO 1 | ARM Instruction Set Architecture

I. Multiple Choice

1. b) Reduced Instruction Set Computing (RISC).
2. b) Thumb 16-bit instructions reduce code size and improve memory efficiency.
3. c) Multiple load/store (LDM/STM) transfer blocks of registers in one instruction.
4. c) Change the program counter to a target address.
5. c) NEON ARM's SIMD extension.

II. True or False

1. TRUE.
2. FALSE Thumb is designed to decrease code size.
3. TRUE.
4. TRUE.
5. TRUE.

III. Scenario-Based Activity

1. Thumb for memory-constrained parts, its 16-bit instructions give much better code density than 32-bit ARM code.
2. LDM/STM (Load/Store Multiple) move whole register sets between memory and the register file in a single instruction, cutting instruction count for the repeated bulk transfers.
3. NEON SIMD processes multiple audio/video samples per instruction (e.g. 8—16-bit values at once), accelerating the vector-heavy computation loops and keeping the streams real-time.